Do not write outside the box

0 3 . 1 The student varied the mass of zinc.

What type of variable is the mass of zinc?

[1 mark]

Tick (✓) **one** box.

Control ☐

Dependent ☐

Independent ☐

Question 3 continues on the next page

Do not write outside the box

Zinc reacts with copper sulfate solution to produce salt **A** and copper.

The word equation for the reaction is:

zinc + copper sulfate → salt **A** + copper

0 3 . 2 What is the name of salt **A**?

[1 mark]

_____

0 3 . 3 The temperature of the solution increases when zinc reacts with copper sulfate solution.

Give **two** other observations that can be made when zinc reacts with copper sulfate solution.

[2 marks]

1 _____

_____

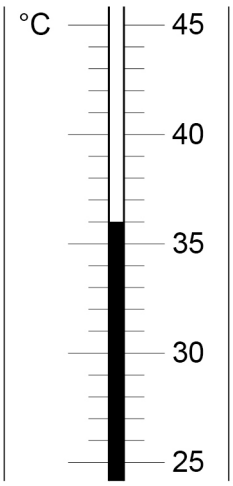
2 _____

_____

Do not write outside the box

0 3 . 4 **Figure 6** shows the highest temperature reached in one experiment.

Figure 6



Complete **Table 3**.

Use **Figure 6**.

[2 marks]

Table 3

Starting temperature in °C	21
Highest temperature reached in °C	
Temperature increase in °C	

Question 3 continues on the next page

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A teacher repeated the investigation using a digital thermometer.

Table 4 shows the results.

Table 4

Mass of zinc in grams	**Temperature increase in °C**			
	Experiment 1	Experiment 2	Experiment 3	**Mean**
1.0	7.8	7.3	7.7	**B**
2.0	13.1	13.8	13.3	13.4
3.0	20.4	12.9	20.2	20.3

0 3 . 5 Calculate value **B** in **Table 4**.

[2 marks]

_____

_____

_____

B = _____ °C

0 3 . 6 The range of the results is the minimum value of temperature increase to the maximum value of temperature increase.

What is the range of temperature increase for the experiment with **2.0 g** of zinc?

[1 mark]

Range = _____ °C to _____ °C

Do not write outside the box

0 3 . 7 One of the results for **3.0 g** of zinc is anomalous.

Which result is anomalous?

Suggest **one** reason why this result is anomalous.

[2 marks]

Anomalous result _____

Reason _____

_____

_____

11

Turn over for the next question